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BSIT 4A

Part C - BIOS vs UEFI Comparison

Feature	BIOS	UEFI
Definition	Basic Input/Output System - firmware that initializes hardware and boots the OS, using 16-bit real mode	Unified Extensible Firmware Interface — modern replacement for BIOS, runs in 32/64-bit mode
Boot Process	Reads the Master Boot Record (MBR) from the first sector of the disk	Uses the GUID Partition Table (GPT) and reads boot files from the EFI System Partition
Max Disk Support	Supports drives up to 2 TB (due to MBR limitations)	Supports drives larger than 2 TB (up to 9.4 ZB theoretically)
Partition Style	MBR (Master Boot Record) — max 4 primary partitions	GPT (GUID Partition Table) — supports up to 128 partitions
Security Features	None built-in	Secure Boot (prevents unauthorized/malicious OS or bootloaders from loading)
Speed	Slower boot — sequential hardware initialization	Faster boot — supports parallel initialization, quicker POST
Advantages	Simple, widely compatible with old hardware/OSes	Faster, more secure, supports modern large drives, has a mouse-friendly GUI, supports network booting
Disadvantages	Slow boot, 2TB disk limit, no security features, outdated	More complex, can have compatibility issues with very old OSes/hardware
Modern Usage	Legacy systems, older PCs, some embedded devices	Standard on virtually all modern PCs, servers, and laptops since ~2012+

Why UEFI Has Largely Replaced BIOS in Modern Computers

UEFI has become the standard firmware interface because it solves several critical limitations of traditional BIOS. First, BIOS operates in 16-bit mode and can only address a limited amount of memory during boot, making it slow and restrictive by modern standards. UEFI runs in 32-bit or 64-bit mode, allowing faster initialization and support for larger, more complex systems.

Second, BIOS relies on the MBR partitioning scheme, which limits disks to 2 TB and only 4 primary partitions — a serious constraint given today's multi-terabyte drives. UEFI uses GPT, removing these limits entirely and supporting up to 128 partitions per disk.

Security is another major factor. UEFI includes Secure Boot, which verifies that only trusted, digitally signed software loads during startup — protecting against bootkits and rootkits.



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BIOS has no equivalent protection.

Finally, UEFI offers a more modern, often graphical interface with mouse support, networking capabilities during boot (useful for remote diagnostics or OS installs), and faster POST (Power-On Self-Test) times. Because of these advantages in speed, security, and scalability, virtually all PCs and servers manufactured in the last decade ship with UEFI as the default firmware, with BIOS now considered legacy technology.